Percents and Percent Change

Concept

Percent means per hundred. Always identify the reference whole.

Convert

Percent ÷ 100 = decimal; 0.4% = 0.004.

Find a part

Part = decimal rate × whole.

Change

Percent change = (new − original)/original × 100%.

Reverse

New = original × (1 ± rate); divide to recover original.

Example 1

Find 0.4% of 75,000.

Solution

1. 0.4% = 0.004.

2. Part = 0.004 × 75,000 = 300.

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Example 2

A price is $54 after a 25% discount. Find its original price.

Solution

1. Remaining fraction = 1 − 0.25 = 0.75.
2. Original = 54/0.75 = $72.
3. Check: 72 × 0.75 = 54.

Example 3

Find the net change after a 20% rise and a 20% fall.

Solution

1. Combined multiplier = 1.20 × 0.80 = 0.96.
2. The final amount is 96% of the original: a 4% decrease.

Percents and Percent Change

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Convert 0.08% to a decimal.

2. Find 240% of 35.

3. 18 is what percent of 48?

4. A value falls from 160 to 124. Find its percent decrease.

5. A price is $69 after a 15% increase. Find the original.

6. Find the net change after a 10% discount followed by another 10% discount.

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Practice answers | check after trying

1. 0.0008.

2. 2.4 × 35 = 84.

3. 18/48 = 0.375 = 37.5%.

4. Decrease = 36; 36/160 = 22.5%.

5. 69/1.15 = $60.

6. 0.90 × 0.90 = 0.81, so the overall discount is 19%, not 20%.